

Laboratory Session: Conditional Statements and Operators in C++

Course: Computer Programming Lab (EE-170L)

Instructor: Rizwan Sir

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Student Name: Hidayat Ullah

University: UET Peshawar

Date: _____

1. Objective

The objective of this lab is:

- To understand comparison (relational) operators
 - To understand logical operators
 - To learn if, if-else, and else-if statements
 - To implement decision making in C++ programs
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2. Theory

Conditional statements are used to make decisions in programming.

Relational Operators

Used to compare two values:

- == Equal to
- Greater than
- < Less than
- > Greater than or equal to
- <= Less than or equal to
- != Not equal to

Logical Operators

- && AND

- || OR
- ! NOT

Conditional Statements

- if → checks condition
 - if-else → checks two conditions
 - else-if → checks multiple conditions
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3. Software Used

- Visual Studio Code
 - MinGW Compiler
 - Windows OS
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4. Task 1: Comparison / Relational Operators

Code

```
#include <iostream>
using namespace std;

int main()
{
    int num1, num2;

    cout << "Enter first number: ";
    cin >> num1;

    cout << "Enter second number: ";
    cin >> num2;

    if(num1 == num2)
        cout << "Numbers are equal";

    if(num1 > num2)
```

```
        cout << "First number is greater";

    if(num1 < num2)
        cout << "First number is smaller";

    return 0;
}
```

Output Example

Enter first number: 10
Enter second number: 5
First number is greater

5. Task 2: Logical Operators

Code

```
#include <iostream>
using namespace std;

int main()
{
    int num;

    cout << "Enter a number: ";
    cin >> num;

    if(num > 0 && num < 100)
        cout << "Number is between 0 and 100";
    else
        cout << "Number is NOT between 0 and 100";

    return 0;
}
```

Output Example

Enter a number: 50
Number is between 0 and 100

6. Task 3: If Statement

Code

```
#include <iostream>
using namespace std;

int main()
{
    int num;

    cout << "Enter a number: ";
    cin >> num;

    if(num > 0)
        cout << "Number is positive";

    return 0;
}
```

Output Example

```
Enter a number: 7
Number is positive
```

7. Task 4: If-Else Statement

Code

```
#include <iostream>
using namespace std;

int main()
{
    int age;

    cout << "Enter your age: ";
    cin >> age;
```

```
if(age >= 18)
    cout << "You are eligible";
else
    cout << "You are not eligible";

return 0;
}
```

Output Example

Enter your age: 20
You are eligible

8. Task 5: Else-If Statement

Code

```
#include <iostream>
using namespace std;

int main()
{
    int num;

    cout << "Enter a number: ";
    cin >> num;

    if(num > 0)
        cout << "Number is positive";
    else if(num < 0)
        cout << "Number is negative";
    else
        cout << "Number is zero";

    return 0;
}
```

Output Example

Enter a number: -5
Number is negative

9. Explanation

- if checks condition
 - else executes when condition is false
 - else-if checks multiple conditions
 - && checks both conditions true
 - Relational operators compare values
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10. Result / Conclusion

In this lab, we successfully learned:

- Relational operators
- Logical operators
- if statement
- if-else statement
- else-if statement